

Program: Diploma in Electrical Engineering & Electric Vehicle Technology	
Course Code: 6417	Course Title: Electric Vehicle Computer Aided Drafting Lab
Semester: 6	Credits: 1.5
Course Category: Program Core	
Periods per week: 3(L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To familiarize with solid primitives and Boolean operations
- To make aware of use of layers and their options
- To identify and apply poly lines and solid modeling options
- To apply layers and their applications, block creation and insertion. Dimensioning
- To identify UCS, VIEW POINT, VIEW PORT, and Multi view of a solid object.
- To appreciate Engineering components in conceptual level and three dimensional level.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic science		Engineering graphics	1
Knowledge of program core course		Basic Automobile Engineering	2
Knowledge of program core course		Automobile Engineering Drawing	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop 3D modeling. Solid primitives and Boolean operations	9	Applying
CO2	Make use of Poly line options to prepare models of electric wiring circuits.	9	Applying
CO3	Build models using Layers, Blocks and Dimensioning tool.	12	Applying

CO4	Make use of various tools and commands for drawing wiring circuits	12	Applying
	Lab Exam	3	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			2			
CO2	3			2			
CO3	3			2			
CO4	3			2			

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop 3D modeling, Solid primitives and Boolean operations		
M1.01	Identify the benefits of 3D modeling.	3	Applying
M1.02	Identify the solid primitives	3	Applying
M1.03	Organize exercises related to applications of 3D primitives and Boolean operations	3	Applying
CO2	Make use of Poly line options to prepare models of electric vehicle wiring circuits		
M2.01	Apply basic uses of poly lines	2	Applying
M2.02	Make use of the options of p edit command	2	Applying
M2.03	Interpret the options in solid modeling.	2	Applying
M2.04	Outline method of creating solid models with extrude, revolve, sweep, loft, press pull	2	Applying
M2.05	Make use of poly lines, p edit and 3D modeling methods	1	Applying
	Series Test–I	1.5	

CO3	Build models using Layers, Blocks and Dimensioning tool.		
M3.01	Identify layers and their applications	2	Applying
M3.02	Apply block creation method	2	Applying
M3.03	Identify electrical symbols, draw control & power circuits of induction motor starters using drafting software.	3	Applying
M3.04	Identify the features of dimensioning	2	Applying
M3.05	Draw the control and power circuits of induction motor starters using drafting software. 1. Direct Online Starter 2. Star Delta Starter	3	Applying
Contents: Layers and its applications. Block creation and its insertion. Dimensioning.			
CO4	Make use of various tools and commands for drawing wiring circuits		
M4.01	Construct wiring of headlight circuit	3	Applying
M4.02	Build the wiring of horn circuit	3	Applying
M4.03	Construct wiring of side indicator	2	Applying

Text/Reference:

T/R	Book Title/Author
T1	AUTOCAD by S.VISHAL, Dhanpat Rai Publishing Company(P)Ltd.
R1	AutoCAD by VISHNUPRIYASINGH, Computer Publishing Limited
R2	MACHINEDRAWING by P.I.VARGHEESE

Online Resources:

Sl. No	Website Link
1	grabcad.com/tutorials/basic-tutorial-for-auto cad
2	https://www.educba.com/category/design/design-tutorials/autocad-tutorial
3	https://www.autodesk.in/campaigns/autocad-tutorials

****Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.

1. Build Automobile components using CAD tools
2. Model the hydraulic brake circuit employing CAD design centre.
3. Develop the sketch of electrical systems using symbols.